

Long-read sequencing as a diagnostic modality for rare disease

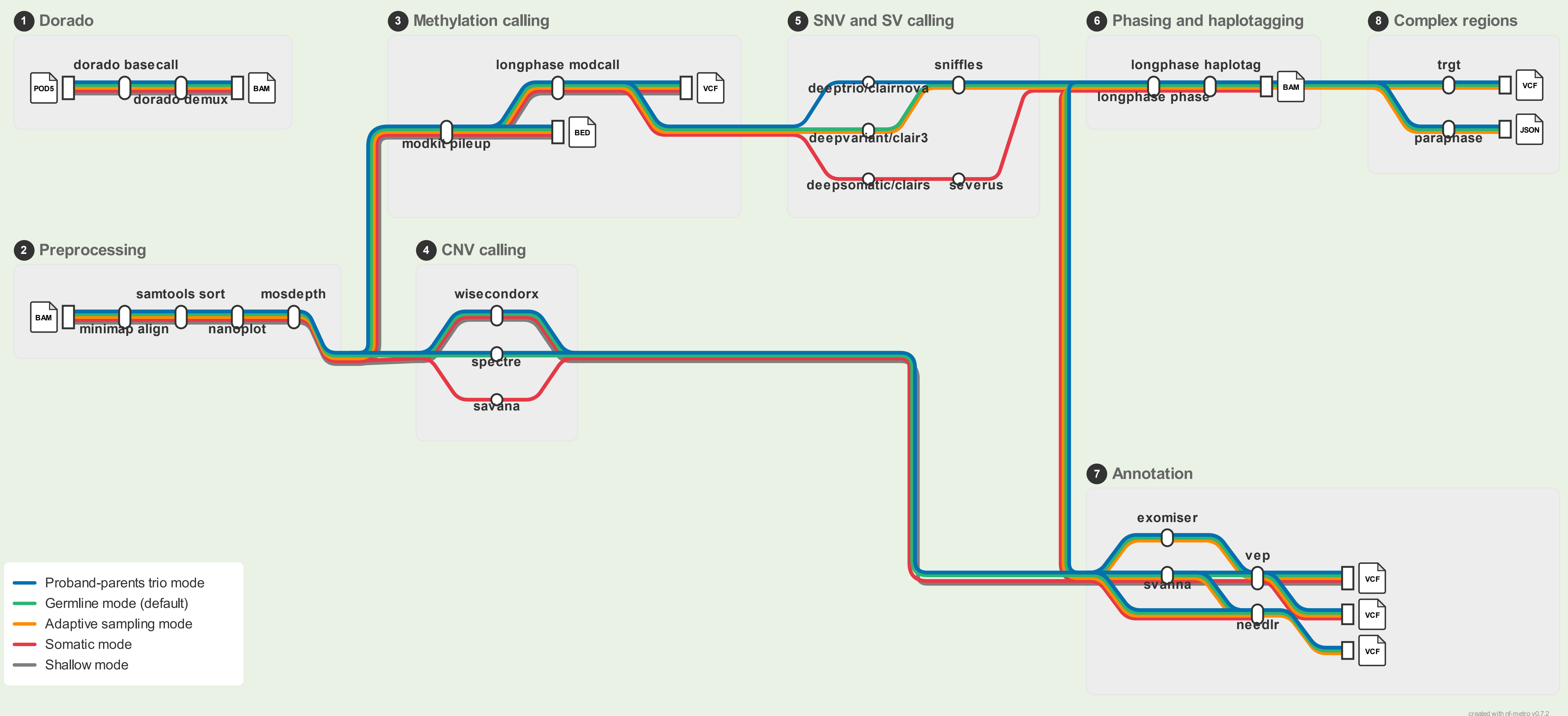
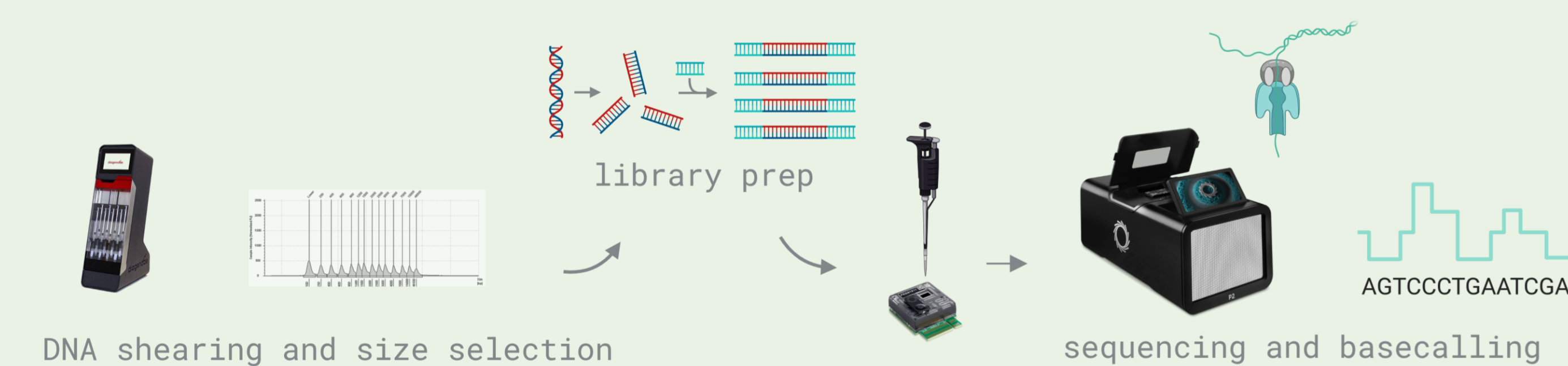
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Background

Approximately 40% of individuals with intellectual disability (ID) remain undiagnosed as short-read sequencing misses cryptic structural variants and variants in complex regions.

Long-read sequencing has demonstrated improved sensitivity for large and complex genomic variation, while also enabling accurate detection of small variants.

Methods



Preliminary results

We successfully resolved a case with a severe neurodevelopmental disorder by phasing variants in the *NFASC* gene.

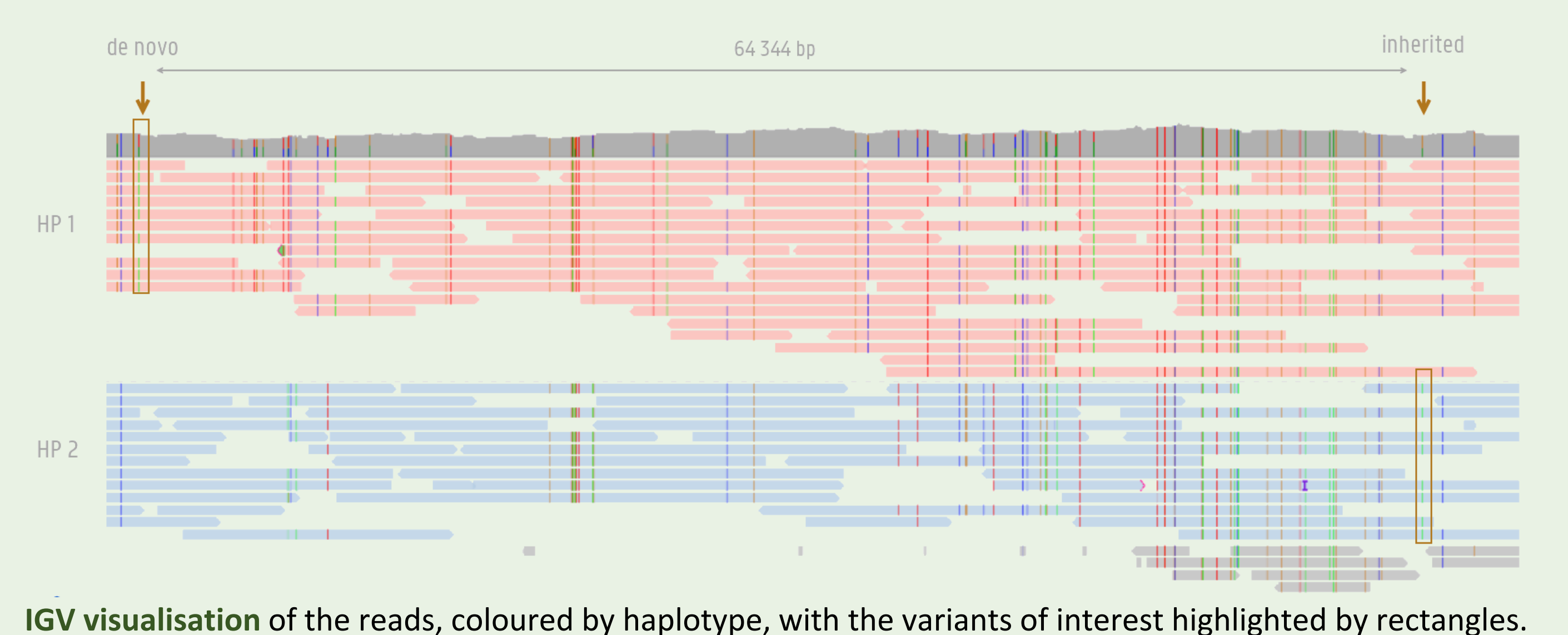
Previously, whole exome sequencing identified two potential causal missense variants in *NFASC*, but we could not confirm their phase.

Using adaptive sampling, we phased the entire gene in a single block, confirming the maternal and de novo variants were in trans.

Eventually, we were able to definitively establish the molecular diagnosis with LRS in this case.

Aims

- **Comprehensive variant detection:** implement a single workflow to identify SNVs (single nucleotide variants), SVs (structural variants), CNVs (copy number variants), tandem repeat expansions, and DNA methylation signatures.
- **Resolve complex genomic regions:** leverage long-read lengths and phasing to disambiguate pseudogenes and determine the exact sequence and length of pathogenic repeat expansions.
- **Determine variant phase:** utilize long-read phasing to establish the relationship between variants.
- **Bridge the diagnostic gap:** target unresolved cases by identifying cryptic variation in regions poorly captured by traditional short-read sequencing.



Future perspectives

- Implementing LRS as a first-tier clinical test.
- Integrating pangenome graphs into the workflow to further enhance SV detection.